Explainable Planning + TOC Bottleneck Analysis Design — v3.27

> **Nature of this document**: Cross-domain methodology paper spanning **Operations Research / Algorithms / ERP / Explainable AI**, describing erpilot v3.27's **Explainable Planning** engine and **Goldratt (1984) Theory of Constraints (TOC)** bottleneck analysis module — design, implementation, and validation.

> 📘 Prerequisites: `MRP_ALGORITHM_DESIGN_EN.md` (v3.25.10) / `MRP_CAPACITY_AWARE_DESIGN_EN.md` (v3.26)

Abstract

While erpilot v3.25.10 and v3.26 achieve operations-research rigor for MRP-II and CLSP heuristics respectively, their outputs remain **black-box** — customers cannot answer fundamental questions like "Why are we pulling so many M6 bolts next week?" This paper describes v3.27's three pillars: (i) **Provenance Graph** [Cheney, Chiticariu & Tan 2009] — tracing upstream causal chain for any planned order; (ii) **TOC Five-Focusing Steps** [Goldratt 1984, 1990] — automatic bottleneck identification from v3.26's CRP load with action suggestions; (iii) **OAT Sensitivity Analysis** [Saltelli et al. 2008] — marginal effect of +20% capacity. We argue these three modules form a **research-grade contribution at the intersection of IE/Algo/ERP/AI**, validated via 12 structural-invariant tests including Schragenheim-Ronen's (1990) 0.85 queueing threshold.

Keywords: Explainable AI, Theory of Constraints, data provenance, sensitivity analysis, ERP

1. Introduction

1.1 The Black-Box Problem

v3.25.10 + v3.26's MRP-II and capacity-aware MRP outputs:

MrpItem (part='M6-BOLT', period='W22')
gross_requirement = 420
net_requirement = 370
planned_order_release = 370

But **no explanation** of why it's 420 and not some other number. Customers (especially non-IE SMB owners) see numbers without context, unable to assess:

- Which SO triggered this 420?
- Did a BOM change affect it?
- How full is our bottleneck work-center?
- If I added 20% capacity, how much better would this plan be?

Vollmann et al. (2005) §11 explicitly states: *"Users will not trust plans they cannot understand."*

1.2 Lessons from Explainable AI

The ML community has well-developed interpretability research [Doshi-Velez & Kim 2017]:

- **LIME** [Ribeiro, Singh & Guestrin 2016, *KDD*] — local linear approximation
- **SHAP** [Lundberg & Lee 2017, *NIPS*] — Shapley value unified framework
- **Counterfactuals** [Wachter, Mittelstadt & Russell 2017] — counterfactual explanation

These target ML models, not IE/OR algorithms, but the **design philosophy** (trace causation, decompose contribution, what-if) applies directly to MRP black-boxes.

1.3 Cross-Domain Contribution

This version's innovation is **first integration of four domains' concepts** in ERP planning:

Domain	Borrowed Concept	Application
IE/OR	TOC five-focusing-steps [Goldratt 1984]; Kingman formula [Schragenheim-Ronen 1990]	Automatic bottleneck identification at 0.85 threshold
Algorithms	Data provenance [Cheney et al. 2009]; DAG traversal	Causal chain tracing
ERP	Complete MPS-MRP-BOM-Routing data chain [Vollmann 2005]	Data foundation for tracing
Explainable AI	OAT sensitivity [Saltelli 2008]; counterfactuals [Wachter 2017]	What-if simulation

2. Methodology

2.1 Module 1: Demand Provenance Graph

Problem formulation: given an `MrpItem` (part i, period t, planned_order_release q), find all upstream events causing q.

Algorithm (reverse DFS, demand upward tracing):

```
ALGORITHM: explain_planned_order(item)
1. Create root node from `item`
2. For each BOMItem b where b.part_id == item.part_id:
3.     parent_product ← b.product_id
4.     parent_item ← MrpItem in same MRP with part_no == product_no
5.     if parent_item exists:
6.         child_node ← recurse(parent_item, depth - 1)
7.         child_node.label += " because of making ..."
8.         root.children.append(child_node)
9. if no parents found:
10.    mps_node ← look up MPS entries for product equivalent
11.    root.children.append(mps_node)
12. return root
```

Complexity: $O(D \cdot F)$ where D = max depth, F = avg fan-in.

Cycle protection: `max_depth` (default 5) caps recursion, preventing infinite recursion from data anomalies.

2.2 Module 2: TOC Bottleneck Analysis (Goldratt)

Five Focusing Steps [Goldratt 1984]:

1. **IDENTIFY** the system's constraint
2. **EXPLOIT** the constraint
3. **SUBORDINATE** everything else to (2)
4. **ELEVATE** the constraint
5. **REPEAT** (if (4) succeeded, find new constraint)

Origin of 0.85 threshold: from **Kingman's formula** [Kingman 1961, *Math Proc Camb Phil Soc* 57], G/G/1 queue mean wait:

$$ W_q \approx \left(\frac{c_a^2 + c_s^2}{2}\right) \cdot \frac{\rho}{1 - \rho} \cdot \tau $$

As $\rho \to 1$, $W_q \to \infty$. Practitioners (Schragenheim & Ronen 1990) find $\rho > 0.85$ triggers sharp W_q increase, hence **0.85 as bottleneck-warning threshold**.

Shadow Price: from LP duality. Under binding constraint, marginal value of +1 unit RHS (capacity). For OR-novice SMB customers, we use minimal approximation: 1.0 when overloaded, 0.0 otherwise.

2.3 Module 3: Counterfactual Sensitivity (Saltelli)

Method (OAT, one-factor-at-a-time):

1. Run baseline plan
2. Modify $C_{kt} \leftarrow \alpha \cdot C_{kt}$ for target WC
3. Re-run Dixon-Silver (other inputs fixed)
4. Compare: overload count / infeasible count / holding cost penalty delta

Limitations (Saltelli et al. 2008, §2.5):

- OAT misses **interactions**; simultaneous multi-factor effects ≠ sum of individuals
- For Global SA, need **Sobol indices** or **Morris elementary effects** (future sprint)

Why still use OAT:

1. SMB customers' intuitive questions are typically single-variable
2. Sobol indices need $N(d+2)$ simulations
3. Saltelli himself: *"OAT in well-understood systems can be informative"*

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3. Implementation

```
backend/app/services/plan_explanation.py     (~500 lines)
├── ExplanationNode (dataclass + render_tree + to_dict)
├── explain_planned_order(db, mrp_item_id, max_depth=5)
├── _explain_via_mps(db, part_id, period, mrp_master_id)
├── BottleneckReport (dataclass)
├── BOTTLENECK_UTILIZATION_THRESHOLD = 0.85    ← Schragenheim-Ronen 1990
├── identify_bottlenecks_from_loads(loads, work_centers)
├── identify_bottlenecks(db, mps_id) → (reports, capacity_result)
├── CounterfactualResult (dataclass)
└── counterfactual_capacity_increase(db, mps_id, wc_id, multiplier=1.2)
```

See `PLANNING_EXPLAINABILITY_DESIGN_ZH.md` §3 for the architecture integration diagram.

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4. Validation

4.1 Structural Invariant Tests (12 cases, all pass)

Test	Validates
`explanation_node_to_dict_roundtrip`	Recursive serialization
`explanation_node_render_tree_shows_hierarchy`	ASCII tree depth indentation
`bottleneck_threshold_at_goldratt_0_85`	Threshold = Schragenheim-Ronen 1990
`bottleneck_identification_basic`	Overloaded WC correctly flagged
`bottleneck_elevation_options_only_when_bottleneck`	No suggestions when not a bottleneck
`bottleneck_elevation_includes_alternate_group_when_set`	alternate_group in options
`bottleneck_shadow_price_positive_at_overload`	LP duality: binding $\Longrightarrow$ shadow > 0
`bottleneck_threshold_boundary`	Strict > 0.85 (not ≥)
`bottleneck_sorted_by_peak_descending`	Reports sorted by peak_util desc
`explain_planned_order_traces_to_mps`	End-to-end: MRP → tree → MPS
`explanation_max_depth_clamps`	max_depth=0 → no recursion
`explain_nonexistent_mrp_item_returns_error_node`	Graceful return on missing

4.2 Results

12/12 tests pass at v3.27 release. Sprint cumulative: **391/391 smoke tests pass.**

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5. Limitations and Future Work

Topic	Why not	Future direction
SO/customer order tracing	Currently starts from MPS; not connected to customer SO	v3.28: MPS_Entry ↔ SO_Item soft link
Multi-factor what-if	OAT limitation	Sobol indices (future research)
Stochastic what-if	Deterministic baseline	Monte Carlo simulation (v3.29 stochastic)
TOC Drum-Buffer-Rope scheduling	Only identification, no scheduling	DBR full impl (v3.28+)
Throughput Accounting	Throughput / OE / I ratios not quantified	Throughput Accounting module
LLM-generated explanation	Currently returns ASCII tree + dict	New explain tool in registry
Provenance edge weighting	Graph treats edges equally	Shapley values for OR (extension of Lundberg-Lee 1.5)

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6. Legal Notice / 法律聲明

⚠️ Important: Legal nature of explanation output

This module produces **explanatory and advisory analysis** built on v3.25.10 + v3.26's IE/OR algorithm outputs. Its results constitute **decision-support information** only — NOT:

1. **Not a legal causation finding**: the "provenance graph" identifies **database lineage at query time**, NOT:
   - Legal causal determinations (e.g., in litigation)
   - Engineering-change responsibility attribution
   - Supply-chain disruption liability attribution
2. **Not a production-decision command**: TOC bottleneck identification uses Goldratt's heuristic framework — outputs are **statistical signals**, NOT:
   - Orders to immediately initiate overtime / outsourcing
   - Requirements for operator scheduling

   - Recommendations for capital expenditure (which requires dedicated financial review)
3. **Not a market-behavior guarantee**: counterfactual sensitivity uses OAT local sensitivity, assuming **other variables unchanged**. In practice:
   - Adding capacity may affect supplier pricing, operator recruitment
   - Multi-factor interactions are unmodeled (see Saltelli 2008 §2.5)
   - Results provide "directional intuition" only — **must not be extrapolated to precise prediction**
4. **LLM wrapper warning** (if enabled in future): when explanations are rewritten by LLM:
   - LLM may hallucinate: produce descriptions inconsistent with raw data
   - LLM may over-simplify: lose important nuance
   - For LLM-rewritten content, **customers should always defer to this module's raw data** (`to_dict()` / `render_tree()`)
5. **Algorithmic limitations**:
   - **Provenance**: based on database snapshot; no time-travel queries
   - **TOC**: 0.85 threshold from G/G/1 queue assumption; may differ for multi-server / batch / priority cases
   - **OAT**: doesn't capture factor interactions [Saltelli 2008]
6. **Disclaimer**: **to the maximum extent permitted by applicable law**, erpilot assumes no responsibility for:
   - **Erroneous accusations** against suppliers/customers/employees based on this explanation
   - **Capital investment errors** due to TOC bottleneck mis-identification
   - **Planning inaccuracy** from counterfactual deviating from reality
   - **LLM rewrite hallucination** consequences
   - **Labor/contractual disputes** with third parties acting on these explanations
7. **Cumulative applicability of predecessor disclaimers**: this version overlays v3.25.10 + v3.26; **all disclaimers** in `MRP_ALGORITHM_DESIGN_EN.md` §6 and `MRP_CAPACITY_AWARE_DESIGN_EN.md` §6 **apply cumulatively** here.

Recommended practice

- Treat provenance trees as "quick data-lookup tools" — not "responsibility reports"
- Have plant manager / production controller review TOC suggestions before overtime / outsourcing / upgrade decisions
- Treat counterfactual results as "directional intuition" — not "precise ROI reports"

- Present LLM-rewritten natural-language explanations alongside raw structured data, allowing cross-verification

7. References

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8. Changelog

Version	Date	Change
v3.25.10	2026-05-20	Uncapacitated MRP-II
v3.26	2026-05-20	Capacity-aware MRP (Dixon-Silver)
v3.27	**2026-05-20**	**This release**: Explainable Planning + TOC + OAT
Future v3.28+	TBD	SO tracing + LLM wrapper + DBR
Future v3.29+	TBD	Sobol global SA + Monte Carlo

Last updated: 2026-05-20 (v3.27) **Authors**: erpilot engineering team (with IE/OR/AI cross-domain academic methodology) **Version**: 1.0 **Chinese version**: `PLANNING_EXPLAINABILITY_DESIGN_ZH.md`